

Role of Backbone Chemistry and Monomer Sequence in Amphiphilic Oligopeptide- and Oligopeptoid-Functionalized PDMS- and PEO-Based Block Copolymers for Marine Antifouling and Fouling Release Coatings

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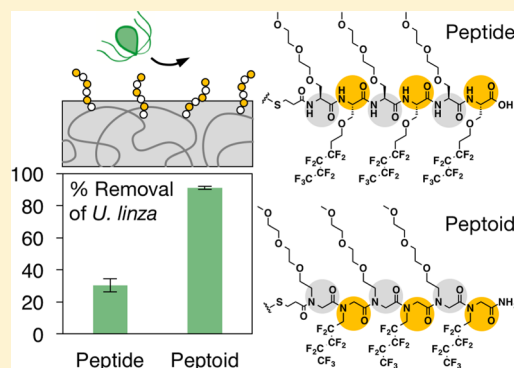
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S Supporting Information

ABSTRACT: Poly(dimethylsiloxane) (PDMS)- and poly(ethylene oxide) (PEO)-based block copolymer coatings functionalized with amphiphilic, surface-active, and sequence-controlled oligomer side chains were studied to directly compare the effects of hydrophilicity, hydrogen bonding, and monomer sequence on antifouling performance. Utilizing a modular coating architecture, structurally similar copolymers were used to make direct and meaningful comparisons. Amphiphilic character was imparted with non-natural oligopeptide and oligopeptoid pendant chains made from oligo-PEO and surface-segregating fluoroalkyl monomer units. Surface analysis revealed rearrangement for all surfaces when moved from vacuum to wet environments. X-ray photoelectron spectroscopy (XPS) spectra indicated that the polymer backbone and oligomer interactions play key roles in the surface presentation. Biofouling assays using the macroalga *Ulva linza* showed that the presence of peptoid side chains facilitated the removal of sporelings from the PDMS block copolymer, with removal matching that of a PDMS elastomer standard. The lack of a hydrogen bond donor in the peptoid backbone likely contributed to the lower adhesion strength of sporelings to these surfaces. Both the initial attachment and adhesion strength of the diatom *Navicula incerta* were lower on the coatings based on PEO than on those based on PDMS. On the PEO coating bearing the blocky peptoid sequence, initial attachment of *N. incerta* showed no measurable cell density.



INTRODUCTION

Marine fouling is caused by the settlement and adhesion of a wide range of marine organisms on surfaces in the marine environment and begins within minutes of immersion with early colonizers such as bacteria, diatoms, and algal spores.¹ When surfaces such as ships' hulls or other manufactured marine structures become fouled, the costs of operation and maintenance are significantly increased.^{2,3} Frequent cleaning adds cost and lowers operational time, and the higher drag penalties induced by a buildup of fouling organisms cause efficiency loss for marine vessels, resulting in increased fuel consumption and greenhouse gas production (representing an extra \$180–260 million per year for the U.S. Navy alone).^{2,4} Paints that contain biocidal compounds have proven successful in reducing biofouling and the associated maintenance and

efficiency penalties;⁵ however, some of these compounds are very persistent in marine environments and have negative effects on nontarget organisms.^{6,7} Environmental concerns have led to increased regulation of these biocidal compounds,⁸ creating a demand for nontoxic alternatives.

Effective coatings should have both antifouling and fouling release properties (reduce the initial settlement of marine organisms and disrupt or weaken the adhesion of those organisms that do attach). Many strategies have been studied extensively, including control of surface topography^{9–11} as well as chemically modifying surfaces using bioactive mole-

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cules,^{12–17} zwitterionic groups,^{18–22} fluoroalkyl groups,^{23,24} and fluoroether networks.^{25,26} Particularly successful coatings have been produced using poly(ethylene oxide) (PEO) polymers and oligomers^{27–31} as well as polydimethylsiloxane (PDMS).^{32–36} Hydrophilic PEO-based block copolymers have shown excellent protein adsorption resistance^{37–39} when compared to hydrophobic materials such as PDMS, while PDMS-based materials exhibit high fouling release due to a low elastic modulus^{40–45} and low surface energy, as described by the Baier curve.^{46–48}

The incorporation of both hydrophilic and hydrophobic components into amphiphilic block copolymers has been shown to produce highly effective antifouling and fouling release coatings.^{30,35,37,49–55} These surfaces incorporate the protein resistance of hydrophilic materials such as PEO with the low surface energy and high fouling release of hydrophobic materials containing siloxane, fluoroalkyl, or alkyl groups. Because of the chemical complexity of these multicomponent materials, the surfaces are environmentally responsive, exhibiting chemical reconstruction underwater.⁵⁵ An effective strategy for incorporating multiple chemical components into a single coating is attaching side chains of various chemistries onto a polymeric backbone. A wide range of amphiphilic side chains have been designed and attached to a variety of polymer backbones to impart amphiphilicity and improved performance.^{31,36,49,52,54,56,57} An advantage to incorporating amphiphilicity via side chains is that structurally similar polymers can be produced, but with modular components that facilitate direct comparisons.

Another advantage to polymer architectures with modular amphiphilic side chains is that the length scale of amphiphilicity can be systematically explored. For chemically heterogeneous coatings, the length scale between different chemical functionalities at the surface is important for antifouling properties, with a critical length scale for low *U. linza* zoospore settlement being on the micrometer scale.⁵⁸ Furthermore, subtle changes in the monomer sequence and spatial distribution of functionalities can have large effects on the chemical surface presentation.^{31,54,59,60} These effects present an opportunity to study how variations in sequence and composition at the molecular length scale can be used to optimize antifouling and fouling release performance, using components that are already in common use.

Surface-active block copolymers (SABCs) have been made with pendant oligomers that are sequence-defined and surface-segregating, produced using peptide and peptoid chemistries, and allow modular incorporation of amphiphilic sequences that can tune the chemical composition of the surface and spatial proximity of functional groups down to the monomer level. Peptoids are peptide isomers, produced via a submonomer approach to solid phase synthesis, in which the residue functionalities are incorporated simply as primary amines and result in an N-substituted glycine repeat unit.⁶¹ Previous work has studied oligopeptides on PDMS block copolymers³⁶ and oligopeptoids on PEO block copolymers,^{31,62,63} all utilizing amphiphilic, surface-active, sequence-defined structures to modify the performance of a polymeric backbone material. In one study with oligopeptoid sequences, fluoroalkyl groups were placed at different positions within the sequence, resulting in dramatically different surface properties.³¹ At least one fluoroalkyl group was necessary to surface segregate the peptoid from within the PEO coating, and positioning fluoroalkyl groups farthest from the block copolymer attach-

ment point produced coatings with the highest surface presentation of peptoid groups. In applying these materials in antifouling coatings, strong trends were seen in *U. linza* settlement with peptoid sequence and in *U. linza* removal with peptoid composition and length. Work with oligopeptides has shown that using very long PEO-like and alkyl side groups is necessary to enhance antifouling and fouling release performance.³⁶ The long side groups, however, eliminate the effects of sequence on the surface properties of the final coatings. Until now, these materials have been studied independently with structures that make direct comparisons difficult.

In this study, structurally similar PEO- and PDMS-based triblock copolymers were synthesized and functionalized with oligopeptide and oligopeptoid side chains (Figure 1) to

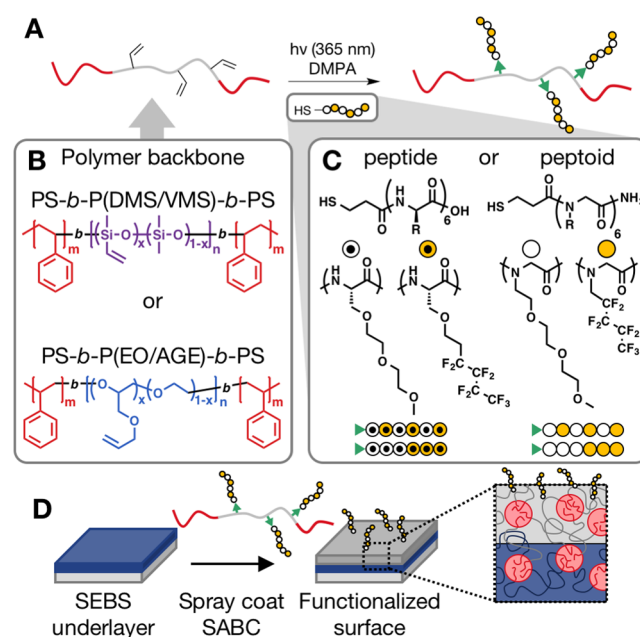


Figure 1. Materials for comparison. (A) Functional oligomers are clicked onto triblock copolymer scaffolds for surface presentation. (B) Polymer backbones: PS-*b*-P(EO/AGE)-*b*-PS and PS-*b*-P(DMS-VMS)-*b*-PS, each with $m = 3–7$ kDa PS end blocks and midblocks with degree of polymerization $n = \sim 1000$ containing $x = 2.5$ mol % vinyl groups. (C) Oligomer structures: peptide and peptoid, each with alternating and blocky sequences. (D) Functionalized surface-active block copolymers (SABCs) are spray-coated onto an SEBS underlayer and annealed to form the final surface.

determine how polymer chemistry, oligomer backbone chemistry, and molecular-scale amphiphilicity independently contribute to the antifouling and fouling release performance of a coating. By making sequence-defined oligomer side chains containing nearly identical subunits, the effects of the backbone chemistry within the side chains could be compared directly. The primary difference between the two is that the peptide backbone is able to both donate and accept hydrogen bonds, whereas the peptoid with a nitrogen-linked functional group has no hydrogen-bond-donating capability. Since the adhesives of many fouling organisms contain a protein component, interactions between the side chains in the coating with marine bioadhesives could be affected by the presence of hydrogen-bonding capabilities. The oligomers contain equal numbers of either PEO-like or fluoroalkyl groups in two different sequences. The fluoroalkyl groups were used not only as the

hydrophobic components of the amphiphilic coating but also as surface-segregating units that define the surface chemistry. Each sequence contained six monomers, with three PEO-like and three fluoroalkyl side chains, and a terminal unit containing a thiol to allow the oligomer to be functionalized onto the block copolymer via thiol–ene “click” chemistry. The sequences studied include one that alternates the hydrophilic and hydrophobic units to produce the smallest length scale of amphiphilicity (“alternating”) and a second with these groups in segments of three units of each type (“blocky”). The final functionalized SABCs were spray coated and annealed onto an underlying SEBS layer to ensure consistent modulus and create a stable coating via physical cross-links at the SEBS/SABC interface.⁶⁴

By analyzing how each side chain performs when functionalized onto either a PEO- or PDMS-based block copolymer backbone, we can understand the individual contribution of each part of the multicomponent coating material. By utilizing a modular and highly defined coating platform, this study directly compares hydrogen bonding capabilities, length scales of amphiphilic groups, and polymeric backbone chemistries. Design rules from studies of well-defined materials such as these can be used to design next-generation antifouling and fouling release coatings.

■ EXPERIMENTAL METHODS

Materials. All materials were purchased from Sigma-Aldrich and used as received, unless otherwise noted. Polystyrene-*block*-poly(ethylene-*ran*-butylene)-*block*-polystyrene (SEBS, MD6945) and SEBS grafted with maleic anhydride (MA-SEBS, FG1901X) were generously provided by Kraton Polymers. Tetrahydrofuran (THF) was dried by stirring over sodium metal and benzophenone, distilled, and degassed by a freeze–pump–thaw process. Styrene for anionic polymerization of PS-*b*-P(DMS/VMS)-*b*-PS was dried by stirring over ground calcium hydride (CaH₂), then distilled, and degassed by a freeze–pump–thaw process. Styrene for chain extension via controlled radical polymerization from a P(EO/AGE) macroinitiator was filtered through an alumina column to remove inhibitor. Hexamethylcyclotrisiloxane (D3), purchased from Gelest, was dried by stirring in benzene over CaH₂. After drying, the benzene was distilled and the D3 was sublimed into a flask containing benzene and dried styrene polymerized with *sec*-butyllithium. The D3 was stirred at room temperature over the polymerization to further dry the D3 until the solution was clear, then the benzene was distilled, and the D3 sublimed into a clean flask. The benzene was then distilled off the D3 to yield dried trimer. 1,3,5-Trivinyl-1,3,5-trimethylcyclotrisiloxane (V3) was dried by stirring over CaH₂ at 40 °C, then distilled, and degassed by a freeze–pump–thaw process. Dichlorodimethylsilane and chlorotrimethylsilane were purified by simple distillation and degassed by bubbling dry nitrogen through the liquid prior to use. Wang resin (100–200 mesh) was purchased from Novabiochem. Dimethylformamide (DMF), diisopropylcarbodiimide, trifluoroacetic acid, and low-loaded (0.20 mmol/g) Rink amide MBHA resin were purchased from Protein Technologies, Inc. 1H,1H-Perfluoropentylamine was purchased from Manchester Organics.

Synthesis of Siloxane Block Copolymer. Siloxane triblock copolymer (polystyrene-*b*-poly(dimethylsiloxane-*co*-vinylmethylsiloxane)-*b*-polystyrene, PS-*b*-P(DMS/VMS)-*b*-PS) was synthesized as previously described and as shown in Figure SI-1.³⁶ In short, dried styrene was polymerized anionically in benzene with the use of *sec*-butyllithium as an initiator. An aliquot was removed for analysis by GPC. The active chain ends were extended with the addition of purified D3 monomer to initiate siloxane polymerization. After 24 h, dry THF was added to accelerate the polymerization of D3. After 2 h, the appropriate amount of V3 was added in THF via syringe pump over 48 h and allowed to react for an additional 2 days. Samples were analyzed via GPC and ¹H NMR to monitor the reaction. The slow

addition of V3 to the copolymerization ensured even distribution of vinyl-containing repeat units along the polymer chain, as V3 is known to polymerize much faster than D3.⁶⁵ At the desired conversion, the active chain ends were coupled with dichlorodimethylsilane in THF and stirred for 16 h before additional coupling agent was added via syringe pump over 24 h to ensure complete coupling. The final triblock was precipitated in methanol, filtered, and dried. The polymer was characterized by GPC to determine final molecular weight, and by ¹H NMR to determine vinyl content (Figures SI-2 and SI-3).

Synthesis of Ethylene Oxide Block Copolymer. The poly(ethylene oxide-*co*-allyl glycidyl ether) (P(EO-*co*-AGE)) midblock was synthesized as described previously.³⁰ In short, ethylene glycol was used as an initiator for the anionic copolymerization of ethylene oxide and allyl glycidyl ether in THF. The statistical copolymer⁶⁶ was terminated with isopropyl alcohol to produce terminal alcohol groups. The molecular weight of the midblock was determined via GPC with PEO standards, and vinyl content was determined by ¹H NMR. The chain ends were converted to initiators with *N*-*tert*-butyl-*O*-[1-[4-(chloromethyl)phenyl]ethyl]-*N*-(2-methyl-1-phenylpropyl)-hydroxylamine (chloromethyl-TIPNO) in the presence of sodium hydride and then used to grow polystyrene end blocks via nitroxide-mediated radical polymerization. The triblock was washed with hexanes to remove autopolymerized polystyrene homopolymer and analyzed by GPC to ensure a monomodal distribution and ¹H NMR to determine polystyrene end block molecular weight (Figures SI-4 and SI-5).

Synthesis of Non-Natural Amino Acids for Oligopeptide Side Chains. Organosulfonates were prepared by reacting 3,3,4,4,5,5,6,6,6-nonafluoro-1-hexanol or diethylene glycol monomethyl ether with methanesulfonyl chloride under nitrogen in THF for 18 h. Excess methanesulfonyl chloride was removed by passing the reaction solution through a plug of silica gel, and the solvent was removed by rotary evaporation.

Non-natural amino acids were synthesized by the nucleophilic displacement of the desired mesylate-bearing side chains by serine-Fmoc in acetonitrile and diisopropylethylamine (DIEA) (Figure SI-6). The reaction was refluxed under nitrogen for 24 h and quenched with DI water. The functionalized amino acids were extracted into ethyl acetate, dried over sodium sulfate, and isolated by removal of solvent by rotary evaporation.

Synthesis of Oligopeptides. The synthesis of oligopeptides via solid-phase synthesis was carried out using published procedures on Wang resin.⁶⁷ Two sequences of hexamer oligopeptides were made: one with three fluorinated amino acids followed by three triethylene glycol amino acids and the other alternating fluorinated and triethylene glycol amino acids. These are referred to as “blocky” and “alternating” oligopeptides, respectively. The N-terminus was capped with 3-mercaptopropionic acid to yield a thiol-terminated oligomer to be attached to a polymer backbone via thiol–ene “click” chemistry. To prevent disulfides from forming, 1.5 equiv of triethylsilane was added as a reducing agent.

Synthesis of 2-[2-(2-Methoxyethoxy)ethoxy]ethanamine for Oligopeptoids. 2-[2-(2-Methoxyethoxy)ethoxy]ethanamine submonomer was synthesized via published procedures.⁶⁸ In short, triethylene glycol monomethyl ether was converted to a mesylate and isolated following procedures outlined above. The mesylate was combined with sodium azide in DMF and stirred at 60 °C for 24 h. The mixture was diluted with excess DI water (taking care to avoid acidic pHs to prevent the formation of toxic and explosive hydrazoic acid), and the azide product was extracted into diethyl ether, washed with water, dried over magnesium sulfate, and concentrated under vacuum. The azide was then reduced with triphenylphosphine in THF under nitrogen overnight. The solid byproducts were removed by filtration, and the solution was washed with toluene and dichloromethane before concentrating *in vacuo*.

Synthesis of Oligopeptoids. The synthesis of oligopeptoids via solid-phase submonomer synthesis was carried out using published procedures using Rink amide MBHA resin.⁶¹ Peptoids were redissolved in 50:50 acetonitrile/water and washed with hexanes before lyophilizing to yield isolated oligomers. Two sequences of

hexamer oligopeptoids were made: one with the incorporation of three 1*H*,1*H*-perfluoropentylamine submonomers followed by three 2-[2-(2-methoxyethoxy)ethoxy]ethanamine (“triethylene glycol submonomer”) submonomers and the other with alternating fluorinated and triethylene glycol submonomers. These will be referred to as “blocky” and “alternating” oligopeptoids, respectively. The N-terminus was capped with 3-mercaptopropionic acid to yield a thiol-terminated oligomer to be attached to a polymer backbone via thiol–ene “click” chemistry.

Thiol–Ene “Click” of Oligomers to Polymer Backbones. Thiol-terminated oligopeptide and oligopeptoid sequences were attached to the siloxane- and PEO-based block copolymers using thiol–ene “click” chemistry. For all cases, 800 mg of triblock copolymer was dissolved in 3 mL of dichloromethane with 1 g of peptide or peptoid and 26 mg of 2,2-dimethoxy-2-phenylacetophenone (DMPA). The reaction solution was first purged with nitrogen and then irradiated with 365 nm UV light from a hand-held UV lamp for 1 h. The functionalized PDMS-based copolymers were precipitated into methanol, filtered, and dried. The functionalized PEO-based copolymers were thoroughly dialyzed in a solution of 15% water in ethanol and dried *in vacuo*. Click completion was determined by ¹H NMR analysis (see Figure SI-7 for representative spectra). A total of eight samples were produced, with peptide and peptoid side chains, two sequences, and attached to two separate block copolymer backbones.

Surface Preparation. Coated glass slides for biofouling assays were prepared as previously reported using SEBS materials.⁶⁴ Briefly, standard microscope glass slides (3 × 1 in.) were treated overnight with freshly prepared piranha solution (5:3 v/v, mixture of concentrated H₂SO₄ and 30 wt % H₂O₂ solution) and then sequentially rinsed with distilled water and anhydrous ethanol. The dried clean glass slides were then immersed in 3.5% 3-(aminopropyl)-trimethoxysilane solution (v/v in anhydrous ethanol) at room temperature overnight, followed by washing with water and anhydrous ethanol. The aminosilane-treated glass slides were cured at 120 °C under reduced pressure for 2 h before slowly cooling to room temperature. The first layer coating was then immediately applied by spin-coating with SEBS/MA-SEBS solution (7% w/v SEBS and 2% w/v MA-SEBS) in toluene (2500 rpm, 30 s), followed by curing at 120 °C in a vacuum oven at reduced pressure for 12 h, allowing the maleic anhydride groups on the polymer backbone to react with amine groups on the glass surfaces, therefore improving the bonding of the coating to the glass. The second layer was spin-coated with SEBS solution (12% w/v SEBS solution) three times (2500 rpm, 30 s), followed by annealing at 120 °C in a vacuum oven at reduced pressure for 12 h to give a base layer thickness about 1 μm. The functionalized PDMS- and PEO-based SABCs were dissolved in 19:1 dichloromethane/toluene and spray-coated onto the surface using a Badger model 250 airbrush to form a coating tens of micrometers thick and annealed in a vacuum oven at reduced pressure at 60 °C for 6 h and then 120 °C for 24 h to ensure complete removal of solvents. The unfunctionalized triblock copolymers served as controls and were spray-coated onto the SEBS underlayer and annealed in the same manner as the functionalized coatings.

¹H NMR. ¹H NMR spectra were recorded using a Varian Gemini 400 MHz spectrometer (Cornell) and a Varian VNMRs 600 MHz spectrometer (UCSB) in CDCl₃ solution.

Gel Permeation Chromatography (GPC). GPC was performed on a Waters ambient temperature GPC with a Waters 1515 isocratic HPLC pump (Cornell) and a Waters e2695 GPC (UCSB), both using THF as an eluent. The instruments are equipped with both a Waters 2414 differential refractive index detector and a Waters 2489 UV–vis detector (Cornell) and a Waters 2998 PDA detector (UCSB). Both instruments were calibrated using polystyrene standards; P(EO/AGE) was analyzed against PEO standards.

Water Contact Angle Measurements. Water contact angle measurements were taken using an NRL contact angle goniometer (ramé-hart model 100-00) using the captive bubble method previously described.^{69,70} Briefly, the surfaces were immersed upside-down in deionized water at room temperature, and an air bubble was trapped

against the immersed surface by releasing it from the tip of a 22-gauge stainless steel needle. For each coating, the water contact angle (measured in the external liquid phase) was measured three times from three separate bubbles at different locations on the surface and averaged. These measurements were taken immediately after immersion in water over a total of 7 days, with measurements taken every 24 h to study the reconstruction of the surfaces. Because the captive bubble method is performed *in situ*, the measurements reflect the operating condition of full immersion.

X-ray Photoelectron Spectroscopy (XPS). Measurements were performed on a Kratos Axis Ultra spectrometer (Kratos Analytical, Manchester, UK) with a monochromatic aluminum Kα X-ray source (1486.6 eV) operating at 225 W under a vacuum of 10^{−8} Torr. Charge compensation was carried out by injection of low-energy electrons into the magnetic lens of the electron spectrometer. High-resolution spectra were recorded at 20 eV pass energy at intervals of 0.05 eV. Survey spectra were recorded at 80 eV pass energy at intervals of 0.5 eV. Samples were measured as-annealed as well as after at least 11 days of undisturbed soaking in Millipore deionized water, after which they were wicked dry of macroscopic water droplets and immediately analyzed. Spectra were calibrated to the major peak (285.00 eV for the aliphatic peak of PS, 286.45 eV for PEO, and 284.38 eV for PDMS),⁷¹ the background was subtracted, and peaks were fit using CasaXPS 2.3.16 software.

Settlement, Growth, and Removal Bioassays of *Ulva linza*. For *Ulva linza* assays, nine slides were equilibrated in 0.22 μm filtered artificial seawater (ASW, Tropic Marin) for 72 h before beginning the assays. Zoospores were released into ASW from mature plants using a standard method⁷² and then suspended in a solution of filtered ASW at a concentration of 6.66 × 10⁵ mL^{−1}. Coated glass slides were placed in the wells of quadriPERM dishes, and 10 mL of the spore suspension was added to each well. Spores were allowed to settle in complete darkness for 45 min. After this time period, all slides were washed gently using filtered ASW. Three of the slides for each material were immediately fixed using 2.5% glutaraldehyde in seawater to determine the spore settlement densities. These slides were analyzed under a Zeiss AxioScop 2 fluorescence microscope using AxioVision 4 image analysis software.⁷³ Counting was performed using an automated program for 30 fields of view of 0.15 mm² per slide.

The remaining six slides were immersed in a nutrient supplemented ASW⁷⁴ and cultured in an illuminated incubator at 18 °C for 7 days. The biomass on the surfaces was measured indirectly by fluorescence of the chlorophyll contained in the plants using a Tecan fluorescence plate reader (GENios Plus); the output was recorded in relative fluorescence units (RFU) as the mean of 70 point fluorescence readings taken from the central portion of each slide.

The strength of sporeling attachment was determined using a water jet with an impact pressure of 70 kPa.⁷⁵ The biomass on the surfaces was again determined using the Tecan plate reader to determine the removal of sporelings.

Attachment and Removal Bioassays of *Navicula incerta*. For *Navicula incerta* assays, six slides were equilibrated in 0.22 μm filtered ASW for 24 h prior to testing. Cultured *N. incerta* cells were diluted to an approximate chlorophyll content of 0.25 μg mL^{−1}. Ten mL of the suspension was added to each of six replicate slides of each coating type and allowed to settle for 2 h. After that time, slides were exposed to 5 min of shaking on an orbital shaker at 60 rpm and rinsed with seawater. Three slides were removed and fixed in 2.5% glutaraldehyde for measurement of initial attachment. The samples were air-dried, and the density of cells attached to the surface was counted on each slide by eye using a fluorescence microscope to illuminate the samples. Counts were made for 15 fields of view (each 0.15 mm²) on each slide.

The strength of diatom attachment was determined for three slides of each coating type using a water channel with a shear stress of 33 Pa.⁷⁶ After exposure to shear stress, samples were fixed and the number of cells remaining was counted, as described above.

Statistical Analysis for Bioassays. For settlement of *U. linza* and initial attachment of *N. incerta*, a one-way analysis of variance was performed to determine if there was a statistically significant difference between groups (*p* < 0.05), followed by a post hoc pairwise Tukey

Table 1. Unfunctionalized Triblock Copolymers and Their Characteristics

sample	$M_n^{a,b}$ (kg/mol)	$\bar{D}^a$	mol % vinyl ^b	DP midblock ^{a,c}
PS- <i>b</i> -P(EO/AGE)- <i>b</i> -PS	3.2–48.4–3.2	1.46	2.1	1100
PS- <i>b</i> -P(DMS/VMS)- <i>b</i> -PS	7.3–68.8–7.3	1.45	2.4	930

^aDetermined via GPC (versus PEO or PS standards). ^bDetermined via ¹H NMR. ^cDP = degree of polymerization.

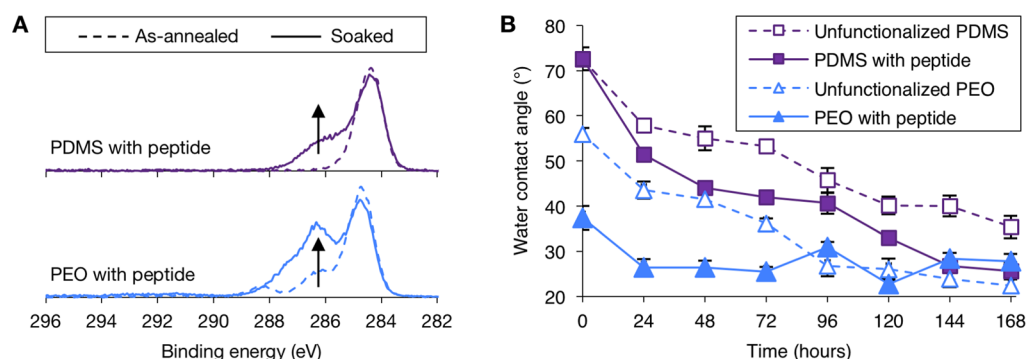


Figure 2. Evidence for rearrangement upon soaking. For ease of presentation, only alternating peptide side chains are shown. (A) High-resolution C 1s XPS spectra of functionalized PEO and PDMS SABCs. Upon soaking, both PDMS- and PEO-based coatings exhibit more C–O bond character (peak at 286.45 eV). (B) Water contact angles via captive bubble contact angle measurements (values reported are measured in the external water phase). All coatings become more hydrophilic over days, with functionalized coatings approaching very similar values regardless of polymer backbone. Error bars show standard deviation of three measurements.

comparison test. For percent removal, these statistical tests were performed on arcsine transformed data of fractional removal of organisms.

RESULTS AND DISCUSSION

Functional surface active block copolymers were synthesized from triblocks of PS and vinyl-containing PEO- or PDMS-based midblocks (Table 1), to which sequence-specific and surface-segregating oligomers (based on peptide or peptoid chemistry) were attached via thiol–ene “click” chemistry. The resulting amphiphilic copolymers were spray-coated onto an SEBS underlayer to form the final antifouling surface. Surfaces were evaluated for chemical presentation via captive bubble water contact angle measurements and X-ray photoelectron spectroscopy and for antifouling and fouling release performance via bioassays of *Ulva linza* and *Navicula incerta*. The key results from comparing polymer backbone material, oligomer side chain chemistry, and oligomer sequence are discussed below.

Polymer Backbone: Comparing PEO- and PDMS-Based Coatings. PEO- and PDMS-based block copolymers have been extensively studied as antifouling and fouling release coating materials, but the diversity of coating chemistries has made it difficult to determine universal design rules. In directly comparing these materials side by side, differences in surface rearrangement (determined by captive bubble water contact angle and XPS measurements) led to differences in fouling performance with the model organisms *Ulva linza* and *Navicula incerta*.

In comparing polymer scaffolds, the relative surface energies of the polymer backbone and side chains define the composition of the surface. In the annealed (dry) state, XPS indicates that PDMS dominates the surface of all PDMS-based coatings (Figure 2A), which also had similar water contact angles equal to that of the unfunctionalized SABC (approximately 71°, Figure 2B). For all PEO-based SABCs, XPS indicates that fluorocarbon-containing oligomers accumulate at the surface in the dry state, and unlike the PDMS-based

coatings, the initial difference between water contact angles of the unfunctionalized PEO coating (56°) and all PEO-based functionalized coatings (32°–37°) was quite large. This difference in initial water contact angle is further evidence that the side chains populate the interface more readily in PEO-based systems than on a siloxane-based backbone, likely due to the low surface energy of the fluorocarbon moieties that drives surface segregation even in the dry state. Conversely, the unfunctionalized PEO SABC surface is enriched in carbon over the theoretical bulk value (89% and 71%, respectively), consistent with an enrichment of carbon-rich polystyrene at the surface in the dry state and a larger initial water contact angle.

Upon exposure to water, surface rearrangement was observed via XPS for all coatings except two (both peptoid sequences on PEO, which consistently display a surface enriched in peptoid side chains). High-resolution C 1s XPS confirms the increase of C–O bonds (286.45 eV) on both PDMS and PEO SABC surfaces (indicated by arrows in Figure 2A). For PDMS-based coatings, this suggests the migration of oligomer side chains to the surface, as C–O ether bonds appear only in the side chains of the PDMS SABC structure and not the polymer backbone. As the coatings hydrated, differences between the water contact angles of functionalized and control materials became more prominent. Within 1 week, all functionalized siloxane SABCs became more hydrophilic than the unfunctionalized siloxane coating and approached similar values in the 26°–30° range. Some factors responsible for these changes are surface rearrangement of functionalized coatings that results in more PEO-like groups at the surface, as well as likely a stable hydration layer forming and water filling the rougher surface created by spray-coating (even the unmodified PDMS SABC becomes more hydrophilic, approaching 35° after 7 days).⁷⁷ The rearrangement of functional groups accounts for the lower contact angles, and thus greater apparent hydrophilicity, of the functionalized coatings as compared to the unfunctionalized PDMS SABC.

For PEO-based coatings, a similar trend of increased hydrophilicity with time was seen. As the coatings hydrated, the contact angles of functionalized and bare PEO samples approached similar values, indicating that under water the PEO and PEO-like side chains gradually dominated the surface.

Overall, all samples became more hydrophilic when immersed in water, with all functionalized coatings approaching similar equilibrium values (in the range of 26° – 31°) distinct from those of either polymer backbone (PEO $22 \pm 2^{\circ}$, PDMS $35 \pm 2^{\circ}$) (Figure 2B). That functionalized coatings based on such dissimilar materials as PEO and PDMS show similar water contact angles indicates that their common peptide and peptoid side chains, not the polymer backbones, define the equilibrium contact angle under water. PDMS-based coatings experience more dramatic rearrangement under hydration, and PEO-based coatings have more consistent side chain presentation regardless of environment. After soaking, the surface presentation of hydrophilic side chains does increase when measured by XPS, but it is difficult to know how closely the structure corresponds to that of the fully hydrated surface due to unknown reconstruction kinetics under vacuum during the measurement, so only qualitative comparisons can be made.

Antifouling and fouling release performance was evaluated using settlement and removal bioassays of the macroalgae *Ulva linza* and the diatom *Navicula incerta*. Initial attachment studies using spores of *U. linza* showed that most samples had similar settlement densities to those of the unfunctionalized PEO and PDMS SABC controls (Figure SI-10). Two samples (alternating peptoid on PEO and alternating peptide on PDMS) outperformed their unfunctionalized SABC controls, indicating possible synergy between polymer backbone and pendant functional groups (Figure 3C).

Exposure of the matured sporelings to a 70 kPa waterjet indicated significant differences in sporeling adhesion strength to the surface depending on polymer backbone (one-way analysis of variance on arcsine transformed data, $F_{9,50} = 36$, $p < 0.05$). In general, the fouling release properties of the PDMS-based coatings were superior to those of the PEO-based coatings (Figure 3D). When functionalized onto the PEO SABC, none of the side chains improved sporeling removal relative to that of the PEO control. On the PDMS SABC, all side chains exhibited fouling release superior to those based on the PEO SABC. In particular, the peptoid side chains on PDMS exhibited much higher removal than both the PDMS control and analogues based on the PEO backbone, with 91–95% of sporelings being removed.

Initial attachment of the diatom *Navicula incerta* was reduced on all the functionalized PEO-based coatings versus the functionalized PDMS-based coatings (one-way analysis of variance, $F_{9,440} = 52.2$, $p < 0.05$). Each functionalized PEO-based coating had significantly reduced initial attachment of diatoms relative to the unfunctionalized control (Figure 3E), with the coating functionalized with a blocky peptoid side chain having zero observed cells attached (Figure SI-11). Except for one coating (blocky peptoid), the addition of side chains to the PDMS backbone did not improve *N. incerta* attachment.

Exposure to a shear stress of 33 Pa indicated significant differences in *N. incerta* adhesion strength depending on polymer backbone (one-way analysis of variance on arcsine transformed data, $F_{9,440} = 133$, $p < 0.05$). In general, the fouling release properties of the PEO-based coatings were superior to those of the PDMS-based coatings, with percent cell removals of 77–89% for PEO-based coatings and 0–12% for PDMS-

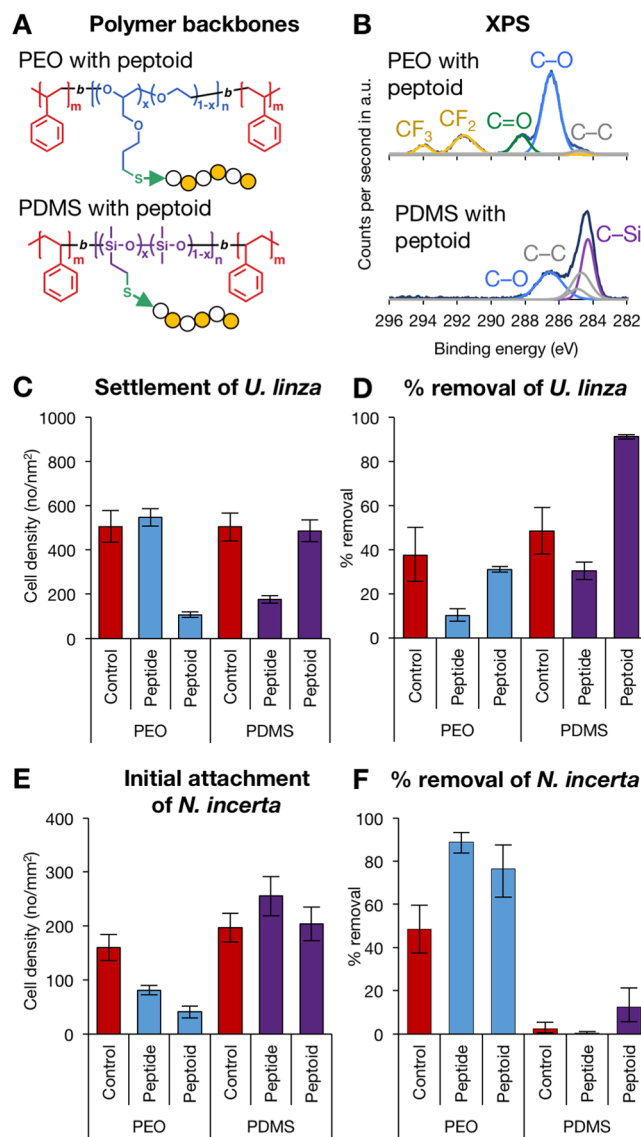


Figure 3. Polymer backbone comparison. For ease of presentation, just results from the alternating sequence are shown. (A) Polymer backbones: PS-*b*-P(EO/AGE)-*b*-PS and PS-*b*-P(DMS-VMS)-*b*-PS, each with 3–7 kDa PS end blocks and midblocks with degree of polymerization $n \sim 1000$ containing $x = 2.5$ mol % vinyl groups. (B) High-resolution C 1s XPS of surfaces soaked for 2 weeks in Millipore water. (C) Settlement results of *Ulva linza* spores. Achieving low settlement depends on a combination of polymer backbone and side chain. (D) Percent removal of *U. linza* sporelings after 1 week of growth with a 70 kPa water jet. PDMS-based SABCs achieve higher removal than PEO-based coatings. (E) Settlement of *N. incerta* cells. PEO-based coatings have lower settlement than PDMS-based coatings. (F) Percent removal of *N. incerta* cells after 30 Pa shear force. PEO-based coatings have higher removal than PDMS-based coatings. For (C), (E), and (F), error bars show 95% confidence limits, and for (D) error bars show standard error of the mean.

based coatings (Figure 3F). The superior performance of PEO-based coatings over PDMS-based coatings is consistent with the findings of previous studies that have shown cells of *N. incerta* tend to be more weakly attached to hydrophilic than to hydrophobic surfaces.^{78–80} The bubble contact angles indicated that the PEO-based coatings had become more hydrophilic than those based on the PDMS SABC after the 24 h of soaking

only the backbone PEO and PEO-like side chains were segregating to the surface. A possible explanation is that the hydrogen-bonding oligopeptides are interacting with each other in the hydrated PEO bulk and either are unable to segregate as strongly to the surface as the oligopeptoids or migrate away from the surface in the XPS due to exposure to air/vacuum on time scales similar to those of the measurement.

In the settlement of spores of *U. linza*, the response to side chain chemistry was dependent on polymer backbone. There was no clear trend with side chain chemistry on PEO SABCs, but peptide-functionalized PDMS coatings had lower spore settlement than their peptoid analogues (Figure 4C). In removal of *U. linza* sporelings, there was also a dependence on polymer backbone, with no clear trend on the PEO SABCs with side chain chemistry, but again a clear difference on the PDMS SABC, with the peptoid samples showing higher removal of sporelings than the peptide samples (Figure 4D). While previous work with oligopeptide-functionalized coatings has been able to increase removal from PDMS,³⁵ the amino acids in that study had longer amphiphilic side groups than those studied here, which likely masked any effect of hydrogen bonding in the peptide backbone.

In the settlement of cells of *N. incerta*, as mentioned above, the PEO-based materials outperformed the PDMS-based coatings, with the peptoid-functionalized coatings having particularly low cell attachment (Figure 4E). There were no measurable cells on the PEO–peptoid coating with a blocky sequence. This superior performance is likely due to the combination of an inherently hydrophilic PEO backbone and an amphiphilic antifouling side chain that lacks hydrogen bonding. In removal of diatoms, there is no trend with side chain (Figure 4F).

Overall, trends with polymer backbone dominated over those with side chain chemistry, but some effects with side chain chemistry stand out. The addition of peptoid side chains significantly enhanced already high-performing polymer backbones: peptoid functionalization of PDMS SABCs improved removal of *U. linza* sporelings, and peptoid functionalization of PEO SABCs reduced *N. incerta* cell attachment. The addition of peptoid side chains also lowered the *N. incerta* settlement on PDMS as compared to the peptide-functionalized analogues. We hypothesize that the superior performance of peptoid side chains stems from the key chemical difference between these otherwise identical oligomers: peptoids lack the ability to donate a hydrogen bond, while peptides have both hydrogen bond donors and acceptors. This difference in intermolecular interactions may be the reason that the peptoid-functionalized coatings outperformed the peptide-functionalized coatings.

Sequence and Length Scale of Amphiphilicity. The main objective in comparing oligomer sequences was to understand how the length scale of amphiphilicity impacts surface presentation and coating performance. Incorporating alternating and blocky sequences of fluoroalkyl and oligo-PEO groups (Figure 5A) resulted in subtle differences in surface chemistry measured with XPS and only affected performance in one bioassay (attachment of *N. incerta*). Overall, sequence effects were less clear than those from side chain or backbone chemistries.

In the dry, annealed state, there was no difference by XPS within pairs of alternating- and blocky-functionalized coatings. After soaking, with the exception of the peptoid-functionalized PEO SABCs, which always present surfaces enriched in peptoid regardless of wet/dry state or sequence, almost all alternating-

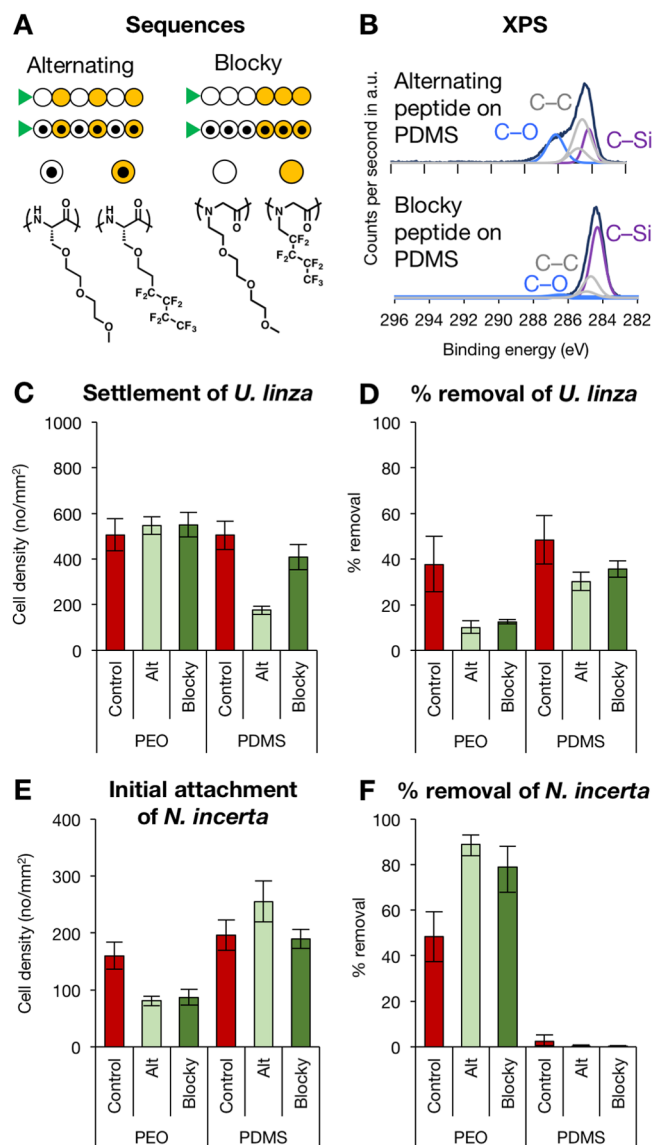


Figure 5. Sequence comparison. For ease of presentation, results for just oligopeptide side chains are shown. (A) Sequences: alternating and blocky hexamers composed of half PEO-like functionalities and half fluorocarbon functionalities. (B) High-resolution C 1s XPS of side chain functionalized surfaces soaked for 2 weeks in Millipore water. For ease of presentation, just peptide-functionalized PDMS SABCs are shown. (C) Settlement results of *Ulva linza* spores. There is no significant trend with sequence. (D) Percent removal of *U. linza* sporelings after 1 week of growth with a 70 kPa water jet. There is no significant trend with sequence. (E) Settlement of *N. incerta* cells. On PDMS SABCs, blocky coatings have lower settlement. (F) Percent removal of *N. incerta* cells after 30 Pa shear force. Removal is dictated by polymer backbone, with no effect of sequence. For (C), (E), and (F), error bars show 95% confidence limits, and for (D) error bars show standard error of the mean.

functionalized coatings showed larger increases in C–O at the surface than the blocky-functionalized coatings (Figure 5B). In a previous study, it was found that surfaces that incorporated oligopeptoids with consecutive (or blocky) fluoroalkyl groups had increased intermolecular interactions.⁶² In the current study, it is possible that the blocky oligomers exhibit similar strong fluorocarbon–fluorocarbon interactions when exposed to air or vacuum, causing the blocky oligomers to become

buried below the surface in the vacuum conditions of the XPS and resulting in spectra very similar to those of the unfunctionalized controls. However, we found that blocky-functionalized coatings performed differently from the controls in fouling tests and achieved the same low water contact angles as their alternating-sequence analogues, indicating that the blocky oligomers are in fact at the surface when in the hydrated state.

For *U. linza* settlement and removal, the sequence of hydrophilic and hydrophobic groups in the side chain made little difference in the efficacy of the coating (Figure 5C), with slightly better performance of alternating sequences over blocky ones. Although a previous study found that increasing the number of consecutive fluoroalkyl groups was worse for both antifouling and fouling release,³¹ the sequences used in this study may not have enough monomer units in each side chain for significant differences to appear. Further studies with longer sequences would likely give further insight into the role of sequence in coating performance.

For initial attachment of *N. incerta*, there was no effect of sequence on PEO SABCs, but PDMS-based coatings functionalized with alternating sequences had statistically higher settlement than their blocky analogues (Figure 5E). There was no difference with sequence in diatom removal (Figure 5F). Overall, differences arising from sequence in surface presentation and antifouling performance were much subtler than differences stemming from side chain or polymer backbone chemistries.

CONCLUSIONS

PEO- and PDMS-based block copolymers have been extensively studied as antifouling and fouling release coating materials, but the diversity of coating chemistries has made universal design rules difficult to determine. The impact of polymer scaffold choice and incorporation of hydrogen-bonding capabilities have been studied independently with different structures, which made direct comparisons difficult. In this study, structurally-similar PEO- and PDMS-based SABCs were functionalized with amphiphilic surface-active oligopeptides or oligopeptoids. Each of these side chain structures incorporated sequence control into a traditional block copolymer architecture and allowed the study of different length scales of amphiphilicity. With this hierarchical and modular design, meaningful conclusions about the roles of subtle functional group differences, surface chemistry design, and environmental response were gathered. By correlating surface characterization with the results from *U. linza* and *N. incerta* biofouling assays, design rules and trends in performance can be used to improve the design of antifouling and fouling release coatings.

After soaking in water, all coatings become more hydrophilic over the time scale of days. All functionalized coatings approach very similar water contact angle values after 7 days, distinct from either unfunctionalized control, indicating a migration of their similar side chains to the surface. PDMS-based coatings have a larger change in water contact angle, correlating to the superior release of *U. linza* sporelings. Furthermore, high-resolution XPS spectra indicate an increase in C–O ether bond character upon soaking for all coatings, corresponding to a migration of ether-bearing side chains to the surface. Peptide-functionalized coatings do not display a fluorine signal after soaking, possibly indicating fast rearrangement kinetics under vacuum measurement conditions that bury peptide oligomers

due to hydrogen-bonding interactions between side chains within the coating.

Fouling release performance was largely determined by polymer backbone chemistry, with *U. linza* having lower adhesion strength on PDMS-based coatings and *N. incerta* having lower adhesion strength on PEO-based coatings. For *U. linza*, peptoid-functionalized PDMS SABCs had the best removal, improved most by peptoid side chains, which had percent removal values of up to 95%. Settlement of *U. linza* spores did not show a clear trend with any of the variables studied, with two coatings showing improved performance (alternating peptoid on PEO SABC and alternating peptide on PDMS SABC). Initial attachment of *N. incerta* was lower on PEO-based coatings, with standout performance from the blocky peptoid PEO SABC, which had zero observed cells.

Differences arising from sequence were much less pronounced than those arising from either polymer or side chain chemistry. Blocky sequences consistently presented less side chain via XPS, indicating possible self-interaction within the coating of consecutive fluoroalkyl groups.

In considering the choice of polymer backbone, this work emphasizes that synergies between these materials can lead to dramatic improvements in performance. Overall, for both fouling species, the peptoid-functionalized coatings outperformed the peptide-functionalized coatings, improving both *U. linza* removal and *N. incerta* settlement, suggesting that coatings devoid of hydrogen bond donors may achieve better overall performance.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.macromol.6b02505.

Detailed synthetic procedures for triblock copolymers, oligopeptides, and oligopeptoids; GPC and ¹H NMR characterization for triblock copolymers and their synthetic precursors; representative AFM and optical micrographs of the SEBS underlayer and spray-coated surface; complete X-ray photoelectron spectroscopy and antifouling/fouling release results for all materials (PDF)

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Notes

The authors declare no competing financial interest.

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